Lecture 6: Forecast Communication, Monitoring, and Governance

Ivy Yang[1]

[1]Economics & Finance, Business School, Southern Illinois University - Edwardsville

From model output to decision communication

Definition: Forecast communication

Forecast communication means translating model output into language, visuals, and action rules that a decision-maker can use.

A forecast is not useful if only the analyst understands it. Managers usually need a short answer to three questions: what is the expected path, how uncertain is it, and what action should we take if the forecast misses? The tradeoff is detail versus usability. A technical appendix may be correct, but the frontline decision often depends on a simple dashboard and a short memo.

Example 6.1. *The notebook builds a manager-facing dashboard covering the January–May 2020 COVID crash window. It shows the champion model's daily S&P 500 return forecasts alongside realized returns, 80% prediction intervals, and risk flags for days where the lower interval bound breaches a drawdown threshold. The dashboard is designed so a non-technical risk manager can immediately identify which days required action—demonstrating that communication is part of the forecasting pipeline, not an afterthought.*

Monitoring forecast performance after deployment

Definition: Monitoring

Monitoring means checking whether forecast accuracy, bias, and uncertainty calibration remain acceptable after the model is deployed.

The environment can change after the model is built, so evaluation cannot stop at the test set. A stable forecasting process tracks rolling RMSE, mean forecast error, and interval coverage over time. The central tradeoff is early detection versus false alarms: tight thresholds catch deterioration sooner, but they can also trigger unnecessary updates.

Example 6.2. *The notebook computes 60-day rolling RMSE, rolling bias, and rolling 80% interval coverage for the champion model deployed from January 2020. The monitoring charts show a*

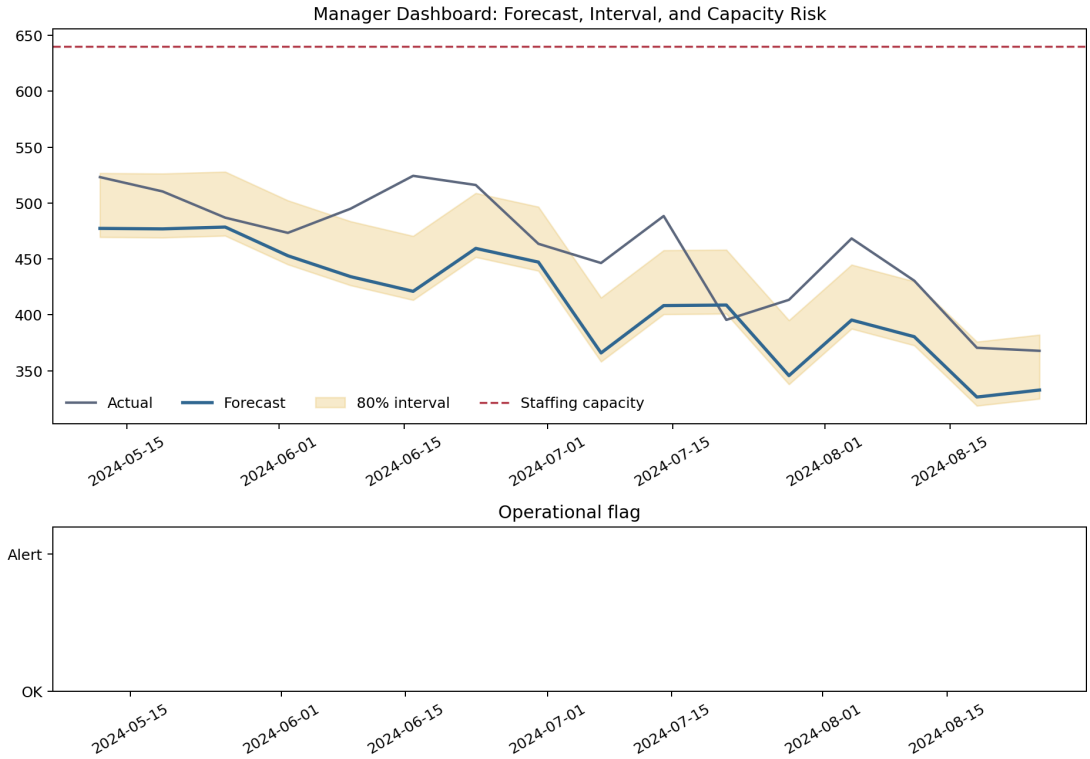

Figure 1: Example 6.1 notebook output: manager-facing dashboard for forecast communication.

sharp deterioration during the COVID crash—rising RMSE, persistent negative bias, and collapsing coverage—well before any scheduled review date. This illustrates why automated monitoring thresholds are necessary: the 2020 failure would have been visible in the metrics within weeks of the crash onset.

Drift and forecast failure signals

Definition: Data drift

Data drift means the distribution of input features shifts away from what the model was trained on, so the model operates in unfamiliar territory.

Definition: Concept drift

Concept drift means the relationship between inputs and the target variable changes, so even correctly observed inputs lead to wrong predictions.

Definition: Failure trigger

A failure trigger is a pre-specified condition that tells the analyst when the model should be reviewed, recalibrated, or replaced.

Figure 2: Example 6.2 notebook output: rolling RMSE, bias, and coverage after deployment.

Two types of drift, two different responses.

Type	What changes	Typical response
Data drift	Input distribution (e.g., VIX spikes into a range never seen during training)	Check data pipeline; consider retraining with recent data
Concept drift	Input-to-output relationship (e.g., VIX's predictive relationship to returns breaks down during a crisis)	Reconsider model structure; collect new training data

Forecast failure is easier to manage when the warning signs are written down in advance. If volatility regimes, cross-asset correlations, or prediction error patterns change sharply, yesterday's model may stop representing today's environment. For governance, the key is to define triggers before the failure occurs. Otherwise, analysts can rationalize bad performance instead of responding systematically.

Example 6.3. *The notebook visualizes the COVID-19 shock as a governance stress test. Four panels show: the VIX distribution in training (2015–2019), during COVID (2020), and post-COVID (2021–2024); the VIX time series with regime shading; the rolling VIX coefficient from*

a re-estimated regression model (concept drift indicator); and the monitoring RMSE rising past its alert threshold. Together they demonstrate that both data drift (VIX values far above training range) and concept drift (changed return sensitivity to VIX) were detectable signals before the worst model failures.

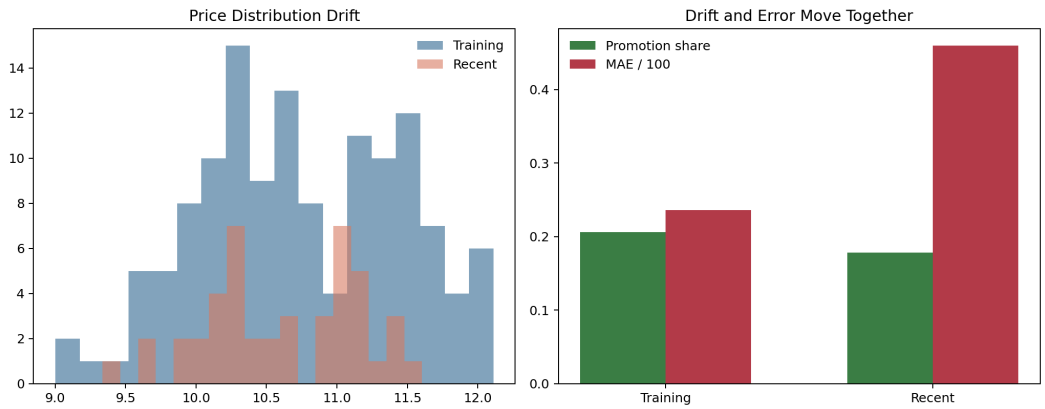

Figure 3: Example 6.3 notebook output: feature drift and rising forecast error after a regime shift.

Champion-challenger evaluation

Definition: Champion-challenger framework

The champion model is the model currently in production; the challenger is a competing model tracked in parallel.

Replacing a production model should not be based on one lucky week. A challenger model should be monitored side by side with the current champion until there is enough evidence that the new model is better on the metrics that matter. The tradeoff is adaptation versus stability: changing models too often creates operational noise, but refusing to compare alternatives can leave the firm stuck with a deteriorating process.

Example 6.4. *The notebook runs a champion–challenger comparison between the Ridge regression (champion, trained on 2015–2019) and a Random Forest (challenger, same training window). Over a 60-day rolling evaluation window in the 2020–2024 deployment period, the notebook tracks which model achieves lower RMSE in each window and plots the rolling RMSE differential. The Random Forest challenger wins a majority of rolling windows—especially during the high-volatility COVID period—providing statistical grounds to consider a model transition.*

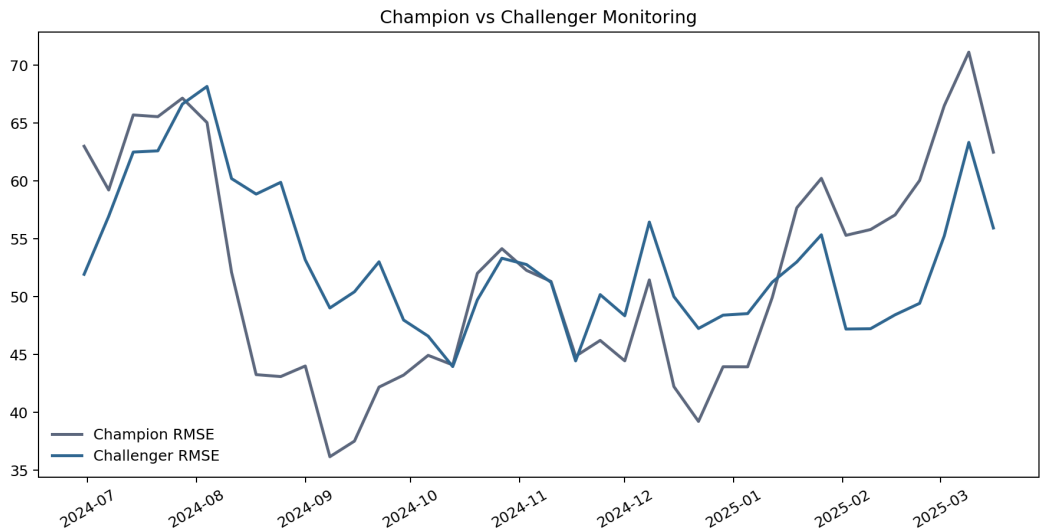

Figure 4: Example 6.4 notebook output: champion versus challenger performance over time.

Overrides, logs, and accountability

Definition: Forecast override

A forecast override is a human adjustment made after seeing the model output, usually because of new business information not present in the data.

Human judgment is sometimes necessary, but it should be governed rather than informal. If a manager overrides the model because of a planned event, supply disruption, or policy change, the reason should be written down and evaluated later. This protects the process from hindsight bias and keeps decision responsibility visible. A good governance system does not eliminate judgment; it records judgment.

Example 6.5. *In the notebook, a simulated override log records five analyst interventions during the COVID monitoring period. Each override is tagged with a date, the model's forecast, the analyst's override value, the realized outcome, and an outcome label (correct or incorrect). The resulting chart shows which overrides improved the forecast and which made it worse—providing a concrete audit trail. This exercise demonstrates why governance requires recording the reasoning at the time of the override, not in hindsight.*

From statistical error to business loss

Definition: Business loss function

A business loss function assigns a dollar cost or operational penalty to each unit of forecast error, making the link between statistical accuracy and organizational impact explicit.

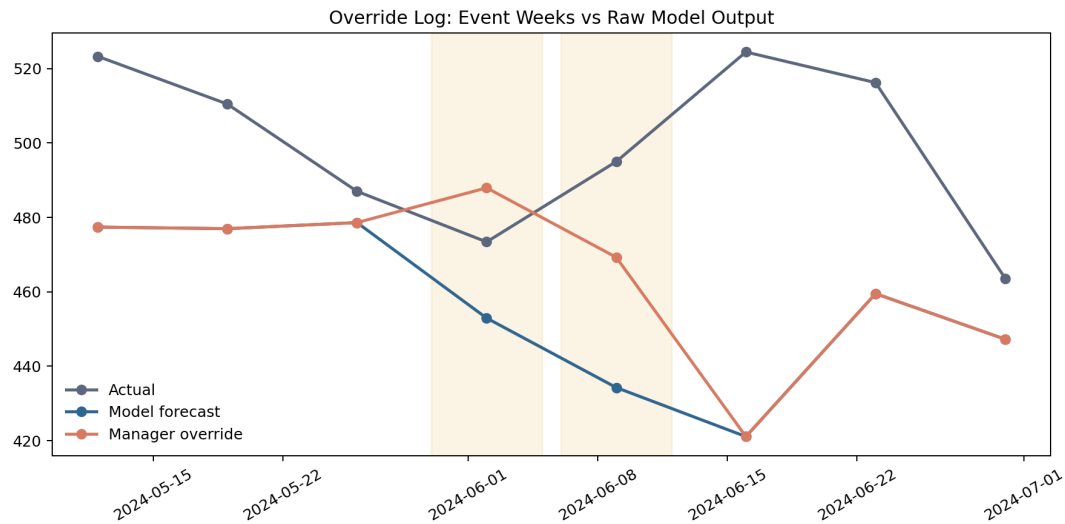

Figure 5: Example 6.5 notebook output: documenting and evaluating forecast overrides.

Statistical metrics such as RMSE and MAE measure error in the units of the target variable. Business impact depends on how those errors translate into cost, risk, or missed opportunity.

Why the translation matters.

- A 5-basis-point forecast miss when a portfolio carries minimal exposure may be negligible.
- The same miss when the portfolio is at maximum leverage on the wrong side of a market move can breach risk limits and trigger forced liquidation.
- Governance conversations should always reference business loss in dollar terms, not just RMSE in return units.

Simple translation framework. Let $e_t = y_t - \hat{y}_{t|t-1}$ be the forecast error at period t. Apply the asymmetric cost structure from Lecture 5:

$$\text{Business Loss}_t = c_{\text{under}} \cdot \max(e_t,\, 0) + c_{\text{over}} \cdot \max(-e_t,\, 0).$$

Average business loss per period is then a governance metric alongside RMSE:

$$\overline{\text{Loss}} = \frac{1}{T} \sum_{t=1}^{T} \text{Business Loss}_t.$$

A rolling window of business loss alongside RMSE gives the deployment monitor two lenses: is the model getting statistically worse, and is the dollar impact growing? These two questions can diverge when error scale is stable but the cost profile has shifted.

Example 6.6. *The notebook applies the asymmetric loss function to a $1,000,000 simulated equity portfolio, comparing the champion Ridge model and the RF challenger over the full monitoring*

period (2020–2024). On each day, the model's forecast error is translated into a dollar cost using $c_{over} = 3$ and $c_{under} = 1$. The 60-day rolling average cost shows that the RF challenger incurs meaningfully lower business loss during the COVID crash window, providing a dollars-and-cents argument for model transition that complements the RMSE comparison from Section 4.

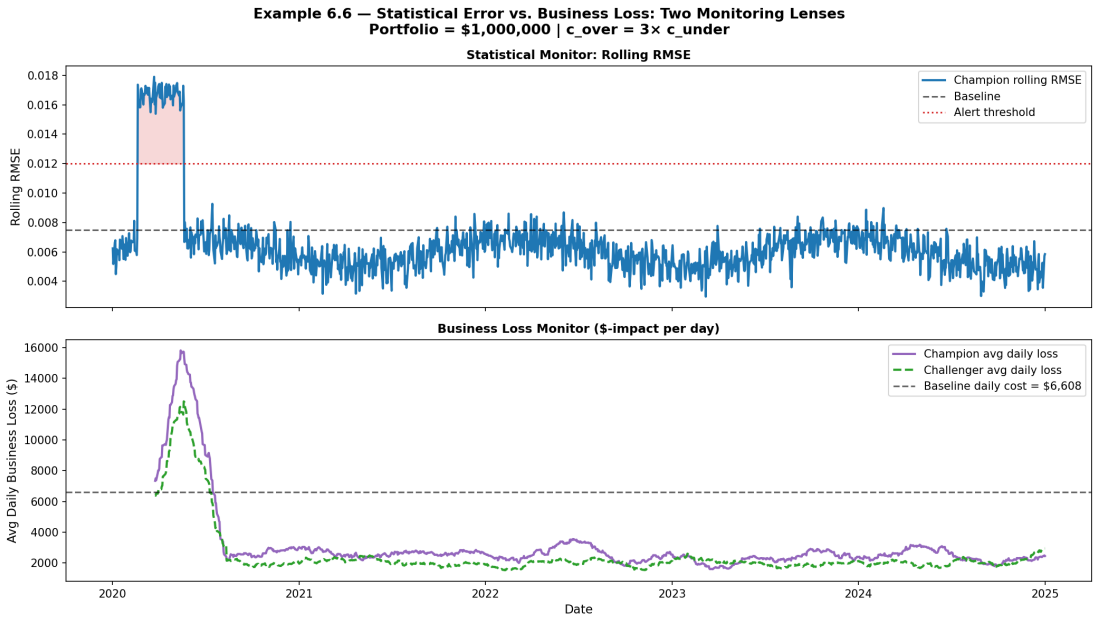

Figure 6: Example 6.6 notebook output: rolling 60-day business loss (champion vs. challenger) alongside rolling RMSE for the $1M equity portfolio.

Conceptual introduction to deep learning for forecasting

Definition: LSTM (Long Short-Term Memory)

An LSTM is a type of recurrent neural network designed to learn patterns from long sequences, capturing dependencies across many time steps without requiring the analyst to manually specify AR orders or interaction terms.

Deep learning methods such as LSTM networks can learn complex temporal patterns directly from data. This is appealing for long, complex time series where the relevant memory depth is unclear or where patterns shift in subtle ways.

How LSTM differs from the models in this course.

Feature	Methods in L1–L6	LSTM
Interpretability	High (coefficients, rules, intervals)	Low (millions of weights)
Data requirement	Works on small samples	Needs large training datasets
Validation discipline	Train / val / test split	Identical—same rules apply
Forecast intervals	Residual quantiles or bootstrap	Requires additional techniques
Governance / auditability	Straightforward	More complex

Why institutions are cautious with deep learning. In regulated settings such as banking, healthcare, and public agencies, model explainability is often required by policy. A model that cannot be explained to a non-technical committee is difficult to deploy, regardless of its predictive accuracy. This is one practical reason why simple, well-validated linear and tree-based models remain dominant in many institutional forecasting workflows.

The key insight is not that LSTM is better or worse: it is that the same validation discipline always applies. Train/validation/test split, out-of-sample RMSE, uncertainty quantification, and monitoring after deployment are all still required, regardless of model complexity.

Example 6.7. *LSTM-style models are commonly used for high-frequency financial data, options pricing, and energy load forecasting at scale. For the daily-return forecasting problems in this course—relatively short series, modest predictor sets—simpler validated models typically perform comparably and are much easier to maintain and explain. The notebook illustrates this trade-off through a complexity–interpretability scatter plot, where each model type (OLS, Ridge, RF, LSTM) is positioned by its data requirement, interpretability, and governance cost.*

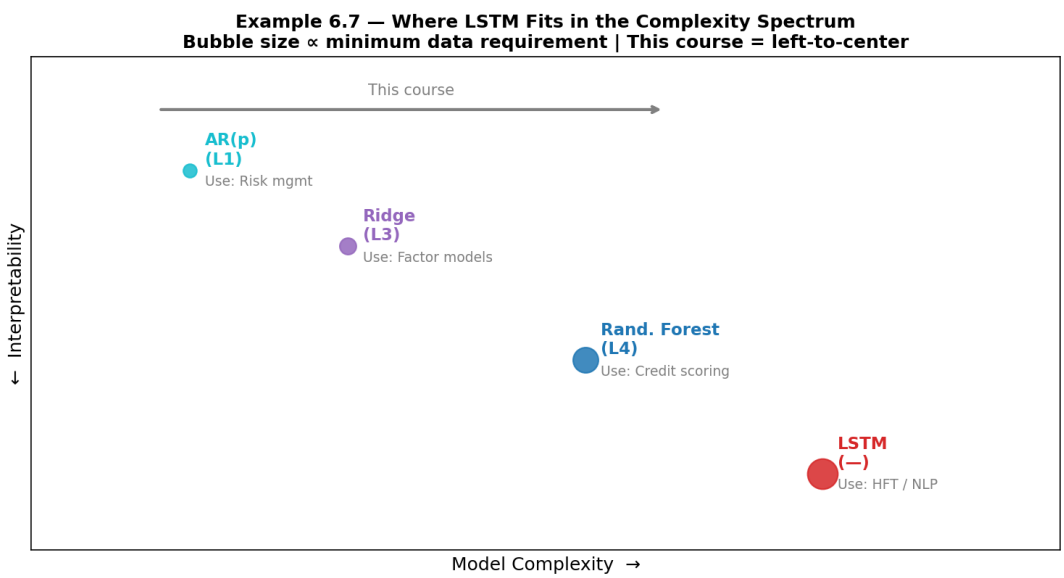

Figure 7: Example 6.7 notebook output: model complexity vs. interpretability positioning, with LSTM shown as high-complexity and low-interpretability relative to methods used in this course.

Practical decision rule for model choice.

1. Start with the simplest model that beats baselines out of sample.
2. Add complexity only when validation results clearly justify it.
3. Use LSTM or other deep learning only when the dataset is large, simpler models leave substantial unexplained structure, and the deployment context tolerates lower interpretability.

Governance checklist for beginner forecasters

Definition: Model governance

Model governance is the set of rules that make forecasting reproducible, auditable, and safe to use in real decisions.

For this course, governance can be kept simple. Before declaring a model ready for use, a forecaster should be able to answer all five of the following questions:

1. **What data were used?** Source, time range, preprocessing steps, and any known data quality issues.
2. **What validation rule was followed?** Chronological split or time-series CV; train/val/test proportions; no look-ahead.
3. **What uncertainty measure was reported?** Interval width, coverage rate, and whether the error distribution was checked.
4. **What monitoring threshold was chosen?** The rolling RMSE or bias level that would trigger a review.
5. **What would trigger re-estimation?** A specific drift signal, a coverage drop, a regime event, or a scheduled calendar date.

A model is not complete until its maintenance and communication plan are also clear. This is the bridge from classroom forecasting to professional forecasting practice.

Example 6.8. *The notebook ends with a governance checklist printout that evaluates the champion model against all five questions above, producing a pass/fail status for each criterion. For a model trained on 2015–2019 data and deployed in January 2020, the checklist reveals specific weaknesses: no pre-specified drift trigger for a VIX regime shift, and no coverage monitoring threshold. This exercise gives students a concrete template they can apply to any forecasting project.*

Interview Questions This Lecture Prepares You To Answer

- **"How do you evaluate forecast quality?"**
  "I evaluate on three dimensions: statistical accuracy (out-of-sample RMSE and MAE), uncertainty calibration (whether prediction intervals cover outcomes at the stated rate), and business impact (whether forecast errors translate to acceptable portfolio cost). Evaluation continues after deployment through rolling monitoring of all three. In our equity return forecasting example, that means tracking daily drawdown flags, not just average RMSE."

- **"Why didn't you use a more complex model?"**
  "Complexity requires justification, not just availability. In risk-regulated settings, simpler models are easier to explain, audit, and maintain. In our context, a validated Ridge or Random Forest model produced acceptable out-of-sample performance, and the governance cost of a black-box model—especially for a risk committee that needs to approve the methodology—was not justified by the marginal accuracy gain. If we had more data and a clear unexplained residual pattern, adding complexity would be worth revisiting."

- **"What would make you change your approach?"**
  "I would revisit the model if rolling RMSE rises significantly above its pre-deployment baseline, bias drifts persistently in one direction, or prediction interval coverage drops below the target rate. I would also reconsider if a confirmed regime shift occurs—such as a COVID-scale volatility spike, a structural break in cross-asset correlations, or a change in data pipeline. In that case, I would first check whether the issue is data drift (VIX range, correlation structure) or concept drift (changed return sensitivities), then decide between recalibration and full re-estimation."